GameWorld: Towards Standardized and Verifiable Evaluation of Multimodal Game Agents

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Research Questions

- How can multimodal large language models be systematically evaluated as game agents across diverse interactive environments with standardized and verifiable metrics?
- What are the comparative performance characteristics of Computer-Use Agents vs. Generalist Multimodal Agents in game-playing tasks?
- What are the key capability bottlenecks preventing current MLLMs from achieving human-level performance in video game environments?

Core Idea

Existing game agent benchmarks suffer from inconsistent evaluation methods, relying on heuristics or VLM-as-judge approaches that introduce noise and make results hard to reproduce. GameWorld solves this by creating a standardized browser-based benchmark with deterministic state-verifiable evaluation - instead of judging success through visual analysis, it directly reads game state variables to compute progress and success metrics. The key insight is decoupling inference latency from gameplay by pausing games during model reasoning, ensuring performance reflects decision quality rather than response speed. Two agent interfaces are compared: Computer-Use Agents that emit raw mouse/keyboard commands, and Generalist Agents that use semantic actions parsed deterministically into low-level controls.

Key Formula

Task Progress Formula

$$\text{progress}_i = \text{clip}_{[0,1]}\left(\frac{q_i^{\max} - b_i}{\tau_i - b_i}\right)$$

Computes normalized progress for run i where q_i^max is the best score achieved in the run, b_i is the starting score, and τ_i is the target score. This measures how far the agent advanced toward the objective within the fixed step budget, providing partial credit for incomplete runs.

Invalid Action Rate

$$\text{IAR} = 1 - \frac{\sum_{r \in R} \#\text{valid actions}(r)}{\sum_{r \in R} \#\text{proposed actions}(r)}$$

Measures instruction-following reliability as the fraction of proposed actions that fail tool-call parsing, role constraints, or parser checks across all runs R. Lower IAR indicates better adherence to action space constraints and output formatting requirements.

Key Figure

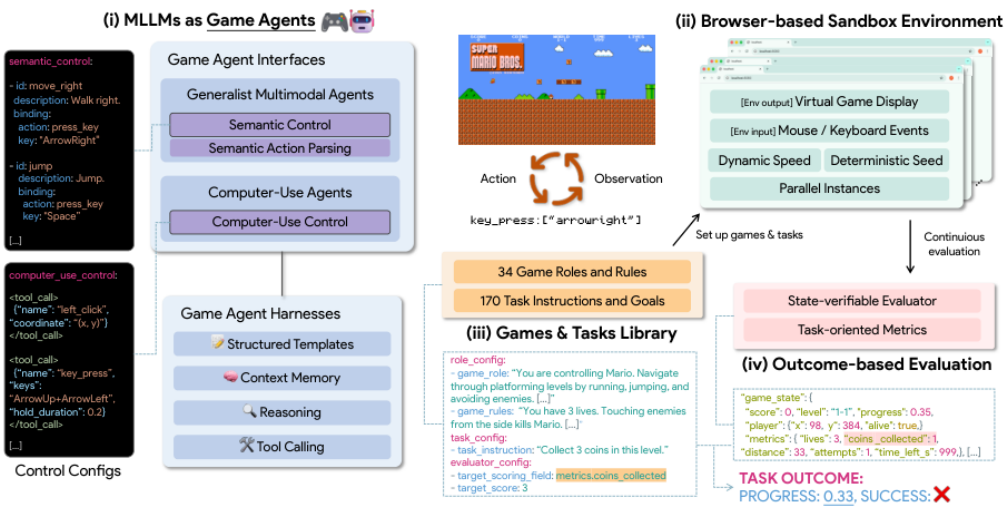

Figure 2 Overview of the **GameWorld** benchmark with four modules: (i) **MLLMs as game agents**, (ii) **Browser-based sandbox environment**, (iii) **Games & tasks library**, and (iv) **Outcome-based state-verifiable evaluation**. This closes a continuous and interactive observation-action-verification loop for systematically evaluating game agents.

Figure 2: Shows the complete GameWorld system architecture with four key modules: MLLM agents, browser sandbox, games library, and state-verifiable evaluation, illustrating how the benchmark closes the observation-action-verification loop.

Limitations

- Limited to browser-based games, excluding native PC games or console titles that might require different interaction paradigms
- Fixed action budget of 100 steps per task may not adequately capture long-horizon planning capabilities in complex simulation games
- Real-time evaluation (GameWorld-RT) shows substantial performance degradation, indicating current models struggle when inference latency affects gameplay
- Human baseline uses only post-graduate students rather than diverse skill levels, potentially underestimating the human-AI performance gap